Use work-energy to solve motion

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Choose work-energy when force and displacement are known.

SUCCESS CRITERIA - I can...

1. List all forces doing work.
2. Calculate the kinetic-energy change.
3. Check whether the result is physically possible.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “work” and “kinetic energy” to explain your answer. Sentence starters are optional.

work: energy transferred by a force along a displacement

kinetic energy: energy associated with motion

final: at the end of the interval; subscript f

01 Warm-up: bring back an earlier idea

A **4 N** force acts forward through **6 m**. **Find work** and **show the force**/displacement directions.

Word help: displacement = change from initial position to final position, with direction

Sentence starter: I found ... [unit]. This value means ... in this situation.

Marzano check - circle: ① ② ③

02 Worked example: explain the prediction

Reference: $W_{net} = ½m(v_f^2 - v_i^2)$.
Constant mass. **Use the displacement** along each force component; stopping removes the initial kinetic energy.

Worked reasoning: A **2 kg** cart starts from rest. Constant net force **4 N** acts through **3 m**, doing **12 J**. Then $½mv^2=$**12** gives $v=\sqrt{}$**12**≈3.46 m/s**.

Your turn: 1. **Sketch a force diagram** and an energy bar diagram for this motion.
2. **Name where time** is unnecessary.

Sentence starter: In my work, I used ... to show ... . I labeled it ... .

Initial **Transfers** **Final**

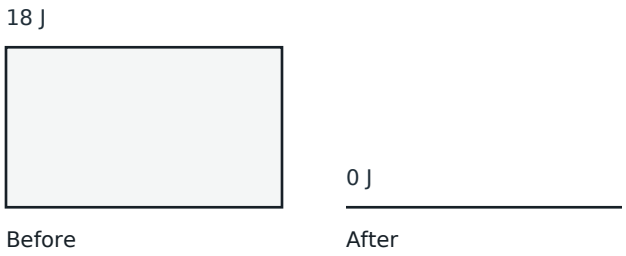
Draw bars or transfer arrows; label forms and joules.

Marzano check - circle: ① ② ③

03 Read and explain the diagram or equation

A cart has **18 J** kinetic energy and then stops. **How much** net work is done? **Show an energy bar before** and after.

Sentence starter: I notice ... . This supports ... because ... .



Bar height represents energy. Read the value and unit above each bar.

Marzano check - circle: ① ② ③

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Continue down each column.

Sentence starters are optional. You may plan in any language.

04 Apply the physics

The cart in question **3** has **18 J** of kinetic energy. A constant **4 N** resistive force brings it to rest.
1. **Show the energy removed** and **calculate the stopping distance**.
2. **Explain why no time measurement** is needed.

Sentence starter: I think … . I can point to …, which shows …, so … .

Initial **Transfers** **Final**

Draw bars or transfer arrows; label forms and joules.

Marzano check - circle: 1 2 3

05 Explain your reasoning

A calculation gives negative final kinetic energy after a long braking distance. What should you check?

Sentence starter: I notice … . This supports … because … .

Marzano check - circle: 1 2 3

06 Review: connect an earlier idea

Two carts approach: one has momentum **6 kg·m/s** *east* and the other **4 kg·m/s** *west*. **Explain why adding their magnitudes** gives the wrong total; **find the signed total**.
Use before and after snapshots to organize the momentum account.

Word help: momentum = mass times velocity, **including direction**

Sentence starter: The step I would change is … . I would change it because I found … .

Before **After**

Label mass, velocity and momentum.

Marzano check - circle: 1 2 3

07 Exit check

A cart has **12 J** initial kinetic energy and gains **8 J** net work.
1. **Find final kinetic energy**.
2. **Use the energy** diagram to **show the quantities you compare**; **label units** and any unknowns.

Sentence starter: Both … . They differ in … because … .

Initial **Transfers** **Final**

Draw bars or transfer arrows; label forms and joules.

Marzano check - circle: 1 2 3

08 STRETCH: test the prediction

A cart needs **18 J** to stop.
1. **Compare stopping distances** for constant opposing forces **3** and **6 N**.
2. **Use the energy** diagram to **show the quantities you compare**; **label units** and any unknowns.

Sentence starter: Both … . They differ in … because … .

Initial **Transfers** **Final**

Draw bars or transfer arrows; label forms and joules.

Marzano check - circle: 1 2 3